

Aviation Operational Performance Analysis Report

Data Analytics Summary (2019–2020) presented by Group 28

Executive Summary

This report analyses airline operational performance across multiple U.S. airports and carriers using arrival flight data, delay causes, cancellations, and diversions. The dataset includes 997,120 arrival flights, of which 170,477 flights were delayed by 15 minutes or more, 9,643 were cancelled, and 1,925 were diverted. The analysis identifies the primary drivers of operational disruptions and highlights areas where airlines and airports can improve efficiency and reliability.

Dataset Overview

The analysis is based on key aviation operational metrics, including:

- Arrival Flights (arr_flights)
- Flights Delayed ≥ 15 Minutes (arr_del15)
- Cancelled Flights (arr_cancelled)
- Diverted Flights (arr_diverted)
- Total Arrival Delay Minutes (arr_delay)

Total delay minutes are attributed to the following causes:

- Late Aircraft Delays => 4,200,591 minutes
- Carrier Delays => 3,826,943 minutes
- NAS (National Airspace System) Delays => 2,505,844 minutes
- Weather Delays => 593,688 minutes
- Security Delays => 18,055 minutes

The analysis evaluates operational performance across airline carriers, airports, and yearly trends between 2019 and 2020.

Key Operational Findings

The results show that late aircraft delays and carrier-related delays account for the majority of total delay minutes. Late aircraft delays are the largest contributor, indicating significant delay propagation across airline schedules, where a late arrival affects subsequent flights scheduled for the same aircraft.

Carrier delays are the second-largest source of disruption and are often linked to internal operational challenges such as crew scheduling, ground operations, and aircraft servicing.

NAS delays also contribute substantially, particularly at high-traffic hub airports where airspace congestion and traffic volume affect scheduling reliability. In contrast, weather delays occur periodically but contribute less overall than operational delays, while security delays remain minimal.

Airport Performance Insights

Airport traffic volume strongly influences operational performance. Major hub airports typically experience higher levels of congestion and operational delays.

For example, Dallas/Fort Worth International Airport recorded approximately 130,007 minutes of delays, reflecting significant airspace congestion and operational demand. Airports with high traffic volumes are more susceptible to delay propagation, as late-arriving aircraft affect gate availability and turnaround operations.

Airline Performance Insights

Airline performance varies based on operational efficiency, scheduling practices, and resource management.

Among the carriers analysed, SkyWest Airlines recorded the highest total delay minutes at approximately 1,804,000 minutes, followed by Southwest Airlines with approximately 1,396,083 minutes. Regional carriers often report higher delay totals because they operate a large number of feeder flights supporting major airline networks.

Key factors influencing airline operational performance include aircraft utilisation efficiency, effective crew scheduling, and real-time operational monitoring.

Strategic Recommendations

To improve operational efficiency and reduce delays, the following strategies are recommended:

- Implement predictive analytics to anticipate disruptions caused by weather and congestion.
- Improve aircraft rotation planning by incorporating schedule buffers to minimise cascading delays.
- Strengthen disruption management procedures to enhance recovery during irregular operations.
- Increase coordination with air traffic management authorities to mitigate congestion in busy airspace corridors.

Business Impact

Enhancing airline operational performance can lead to improved passenger satisfaction, reduced operational costs, increased aircraft utilisation, and stronger brand reputation. Even small reductions in delay minutes can generate significant operational efficiencies and financial savings across airline networks.

Conclusion

The analysis indicates that most aviation delays stem from operational factors rather than environmental conditions. Late aircraft and carrier-related delays represent the largest contributors to overall disruptions. Airlines that adopt data-driven operational strategies, predictive planning, and improved scheduling practices can significantly enhance on-time performance and network reliability.